AI-guided cellular rejuvenation

A proposed program for causal mechanism discovery and functional validation in adult human cells

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A defined first asset in human skin

Initial biological system	Aged human skin cells and reconstructed skin. Tissue repair is a provisional therapeutic direction.
First asset to establish	A controllable intervention that restores independent functional measures while preserving cell identity.
Why this scope	An accessible tissue model permits repeat measurement, perturbation and direct assessment of repair function.
Long-term ambition	Comprehensive reversal of cellular biological aging requires further evidence, broader tissue work and additional capital.

White paper pp. 1–3, 10, 16–17

Current evidence is computational

34

public-data samples

3

donors in the pilot

6

synthetic benchmark settings

The reported AUROC of 1.00 measures treatment classification. It does not measure restored function or biological rejuvenation.

Prospective biology must establish efficacy, a validated target and a usable safety window. No clinical result or lifespan extension is claimed.

AI priorities face prospective biological tests

Cell state	Mechanism	Selection	Validation
Multimodal omics and imaging	Regulatory sequence, perturbation and causal models	Bayesian experiment design under a fixed budget	Orthogonal tests, rescue and independent function

Model value is measured by independently confirmed discoveries per dollar against expert nomination and random eligible targets.

A balanced screen and independent confirmation

60

AI-selected targets

60

expert-selected targets

60

random eligible targets

180 targets in the planned discovery screen across eight adult donors, with comparable experimental budgets and shared controls.

Confirm selected mechanisms in 16 new donors across two laboratories. Require orthogonal perturbation and rescue, functional benefit, preserved identity, safety and persistence after withdrawal.

Biological clocks support the assessment. Functional recovery governs advancement.

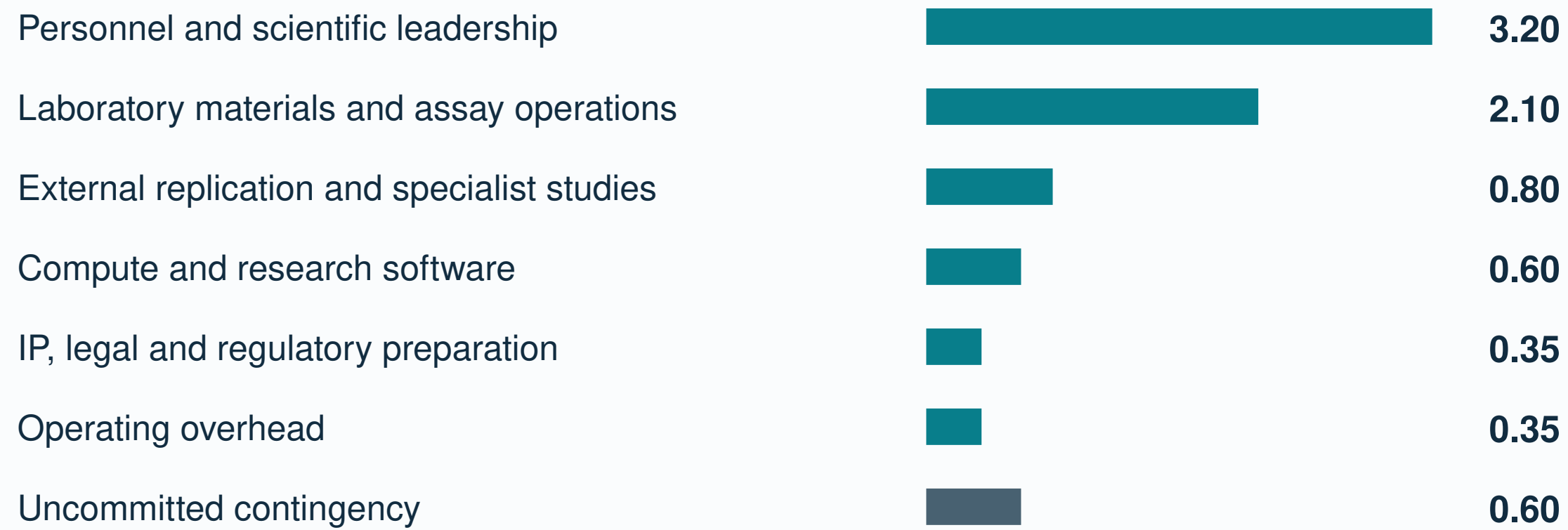
White paper pp. 10–13

Evidence gates control later spending

Months 0–3	Rights review, experimental lead, replayed pilots and qualified assays
Months 4–6	Balanced discovery screen and target engagement. Advance up to 12 candidates
Months 7–12	At least two mechanisms with orthogonal evidence and functional benefit
Months 13–18	Replication in new donors and an external laboratory, with withdrawal follow-up
Months 19–24	One mechanism package with modality feasibility, IP assessment and a next-study plan

White paper pp. 12–13, 23

Illustrative $8 million plan over 24 months



$7.40m operating work + $0.60m reserve = $8.00m

Zero commercial receipts assumed. Clinical trials, GMP and full IND-enabling work require later capital.

White paper p. 18. USD millions. Planning assumptions, subject to quotes and approved hiring

Commercial value requires a reproducible mechanism

Proposed first route	A defined research collaboration or mechanism option with a biotech or pharmaceutical partner.
Asset for partner diligence	Replicated functional benefit, delivery feasibility, a defined development path and assignable rights.
Potential data advantage	A longitudinal record linking interventions to function, identity, safety and failed experiments across donors.
Commercial uncertainty	Public models are widely available. Buyers, pricing, exclusivity and procurement appetite require validation. No signed collaboration is claimed.

White paper pp. 15–17, 19, 21

Major failure risks have explicit decisions

Clock changes alone	Use independent functional endpoints and control for proliferation or selection. Reject candidates without reproducible functional benefit.
Identity loss or safety	Assess cell identity, genome integrity and oncogenic signals. Stop candidates without an acceptable window.
Failure to replicate	Require new-donor and external-laboratory confirmation. Narrow or discontinue nonreplicating programs.
Rights or cost constraints	Establish commercial-use rights before commitments. A 30% rise in lab and external-study costs exceeds the reserve by $0.27m.

White paper pp. 13–14, 18–22

Company formation and first investment discussion

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Formal roles and time commitments remain to be documented. Experienced experimental and translational leadership must be secured.

Incorporation and laboratory base remain open. No employer or institution is represented as a venture sponsor.

We seek an early investment and company-building partner to assess scientific fit and agree the evidence needed for the first financing.